

I. Recap

Hypothesis testing + C.I for β_0 + β_1

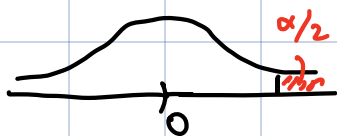
Hyp tests: If $p\text{-value} < \alpha \Rightarrow \text{reject } H_0$

In most cases $H_0: \begin{matrix} \beta_0 = 0 \\ \beta_1 = 0 \end{matrix} \rightarrow H$ we reject H_0 .

then X is a significant predictor for y .

C.I.

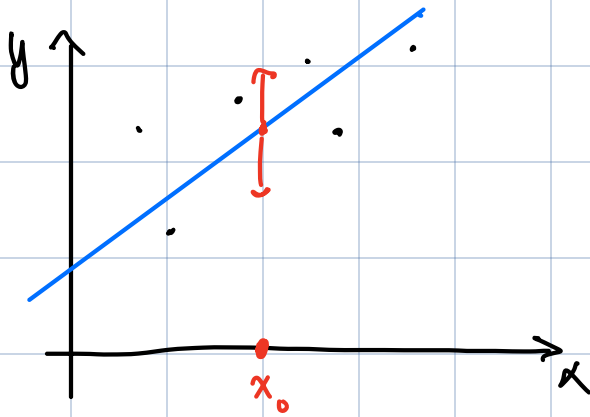
$$\hat{\beta}_1 \pm t_{1-\alpha/2, n-2} \frac{\hat{\sigma}}{\sqrt{SSX}} = (\text{lower}, \text{upper})$$



What if $\emptyset$ is in (L, U) ? Fail to reject H_0 .
 ") not in (L, U) ? Reject H_0 .

Confidence Interval for $E(y)$ (mean response)

Interval s.t. the true mean of y , for $X = x_0$, is in the interval with $(1-\alpha)100\%$ confidence.



$$\hat{y}(x_0) \pm t_{1-\alpha/2, n-2} \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{SSX}}$$

Prediction Interval

Interval s.t. the next observation of y , y_{new} , for $X = x_0$, will be in the interval with $(1-\alpha)100\%$ confidence.

Pred interval for y_{new} when $X = x_0$:

$$\hat{y}_{\text{new}}(x_0) \pm t_{1-\alpha/2, n-2} \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{SSX}}$$

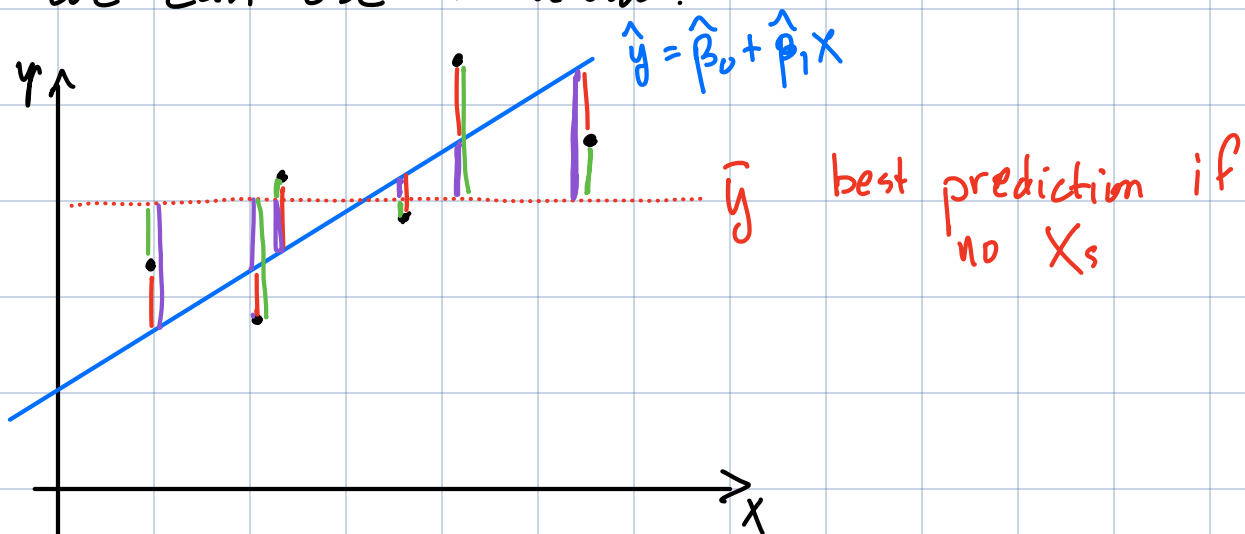
$$\sigma^2 + \sigma^2 \left(\frac{1}{n} + \frac{(x - \bar{x})^2}{SSX} \right)$$

$\hat{y}$

$\epsilon_i \sim N(0, \sigma^2)$

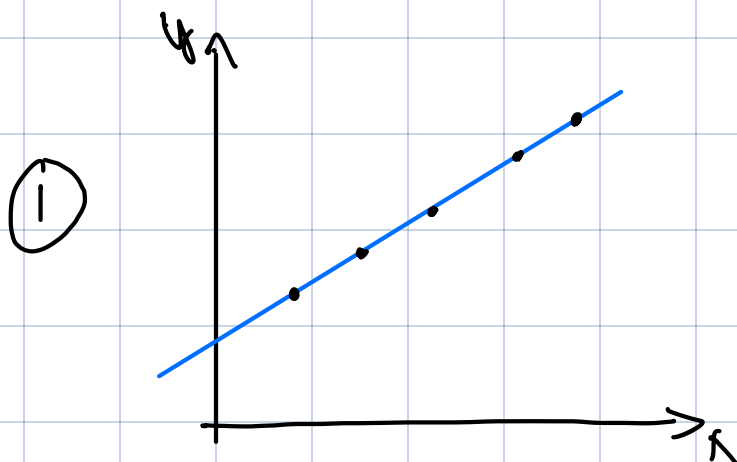
III Sum of Squares Decomposition

To check if X is a significant predictor, we can use residuals.



$$\overset{SSE}{\sum_i (y_i - \hat{y}_i)^2} + \overset{SSR \text{ (regression)}}{\sum_i (\hat{y}_i - \bar{y})^2} = \overset{SST \text{ (total)}}{\sum_i (y_i - \bar{y})^2}$$

Let's look at an extreme case



$$\sum_i (y_i - \hat{y}_i)^2 = 0 = SSE$$

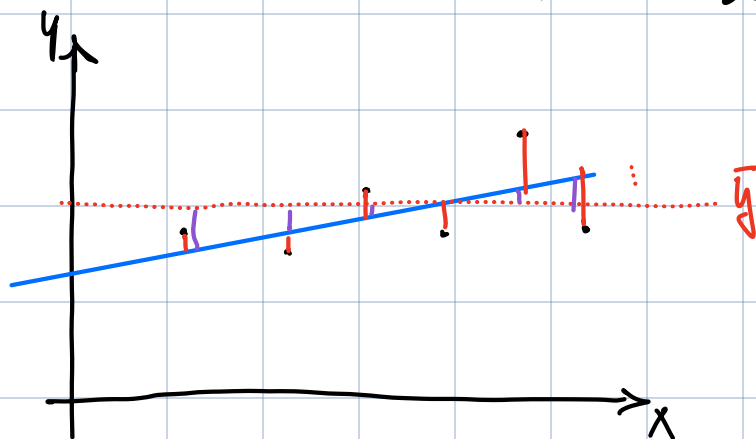
$$\sum_i (\hat{y}_i - \bar{y})^2 = \sum_i (y_i - \bar{y})^2$$

since $y_i = \hat{y}_i \quad \forall i$

Reject $H_0, \beta_1 \neq 0$, SST is made up of

a "Lot More" SSR than SSE.

(2)



Fail to reject H_0 . SST is made up of relatively more SSE than SSR

IV F-test

$$\frac{SSR/1}{SSE/n-2} \sim F_{1, n-2}$$

$\nwarrow p-1$
 $\nwarrow n-p$

$p = \# \text{ of } \beta\text{'s}$

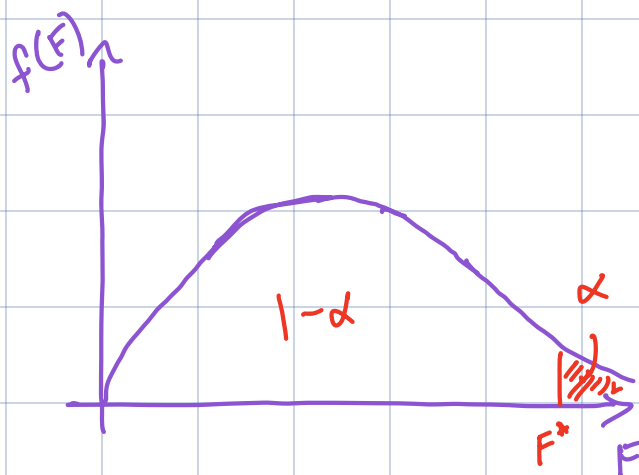
F-distn:

Let $U \sim \chi^2_k$

$V \sim \chi^2_m$

U, V independent

then $F = \frac{U/k}{V/m} \sim F_{k, m}$



$$\frac{SSR}{\sigma^2} \sim \chi^2_1$$

$$\frac{SSE}{\sigma^2} \sim \chi_{n-2}^2$$

$$S_0 \quad F = \frac{SSR/1}{SSE/n-2} \sim F_{1, n-2}$$

What if SSE is large, relative to SSR ? slope not significant
 " " small, " " ? slope is significant

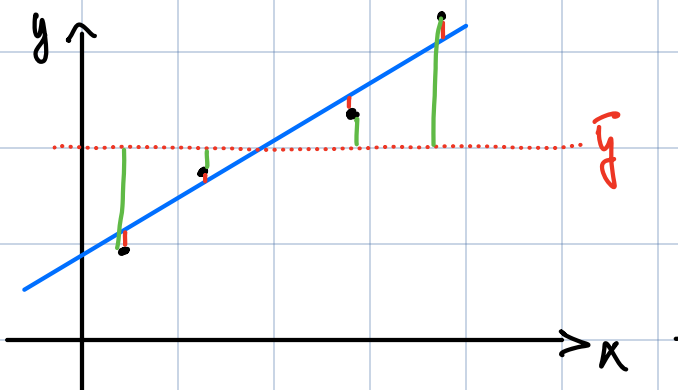
Decision Rule

If $F > F_{1-\alpha, 1, n-2}^*$, reject H_0 and say

"X is a significant predictor of y"

IV Coefficient of Determination (R^2)

$$R^2 := 1 - \frac{SSE}{SST}$$



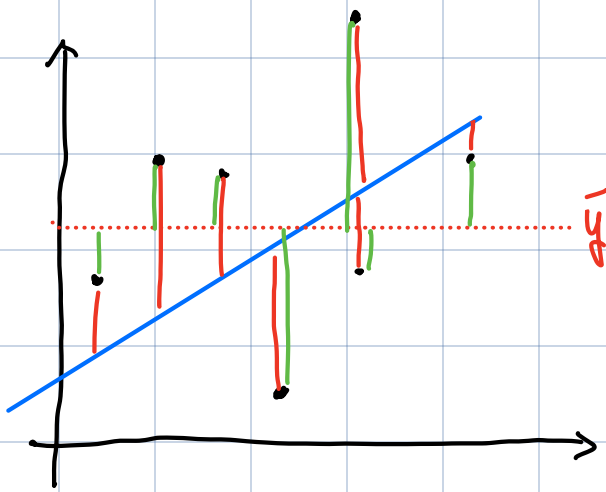
Notice $SSE \ll SST$

$\Rightarrow R^2 \approx 1$ - something small

$$\Rightarrow R^2 \approx 1$$

$$0 \leq R^2 \leq 1$$

Proportion of variance in y explained by the model.



Notice $SSE \approx SST$

R^2 : 1 - something close to 1

$\Rightarrow R^2 \approx$ closer to ①

For SLR $R^2 = \rho^2$ where ρ is the sample correlation coefficient between X and y

$$\rho = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}}$$